

Project Exhibition - 1

Project Title

Smart Digital Complaint Management and Public Transparency System

Project Team

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Short Project Description

The Smart Digital Complaint Management and Public Transparency System is a web-based platform that enables citizens to report civic issues such as road damage, water leakage, garbage accumulation, and faulty street lights. The system provides structured complaint submission, automatic ticket generation, complaint tracking, administrative workflow management, public transparency through live ticket status, and comprehensive report generation after resolution. It aims to improve communication between citizens and local authorities while ensuring accountability and transparency throughout the complaint lifecycle.

Detailed Project Description

The Smart Digital Complaint Management and Public Transparency System is a comprehensive web application designed to digitize the process of reporting, managing, tracking, and resolving civic infrastructure complaints. The platform provides an organized communication channel between citizens and municipal authorities while maintaining complete transparency regarding the status of every complaint.

Citizens frequently encounter public infrastructure issues such as damaged roads, water leakages, overflowing garbage, malfunctioning street lights, broken drainage systems, and other civic problems. In many cases, reporting these issues requires visiting government offices, making repeated phone calls, or relying on inefficient manual processes. Furthermore, citizens often receive little or no information regarding the progress of their complaint.

This system addresses these challenges by providing a centralized online complaint management portal where complaints can be submitted, tracked, reviewed, assigned, and resolved in a structured manner.

Citizen Module

The complaint submission process is designed to collect standardized information so that municipal authorities receive complete and relevant details for each issue.

The citizen begins by selecting the appropriate complaint category, such as:

- Road Damage
- Water Leakage
- Garbage
- Street Light

Based on the selected category, the system dynamically displays a set of category-specific Yes/No questions to gather additional information. These questions help authorities understand the urgency and context of the complaint. Examples include:

- Has there been any accident because of this issue?
- Has the problem existed for more than one week?
- Is a school or hospital located nearby?
- Is there a market or busy public area nearby?
- Is the issue affecting vehicle movement?
- Is the issue causing public inconvenience?

After answering these questions, the user provides:

- Exact location of the issue
- Photographs of the problem
- A detailed textual description
- Name
- Email address

OTP Verification Feature

To ensure the authenticity of the complaint and prevent spam, the system implements a verification step. Before the citizen can submit the ticket, they must request an OTP (One-Time Password) which is instantly dispatched to their registered email address. The citizen must enter this OTP on the submission page. The ticket can only be submitted once the system successfully validates the OTP.

Once validated and submitted, the system automatically generates a unique Tracking ID and sends a confirmation email containing:

- Complaint details
- Complaint category
- Tracking ID
- Date and time of submission

The citizen can use this Tracking ID to monitor the progress of the complaint throughout its lifecycle.

Complaint Tracking

Every complaint passes through predefined stages, allowing citizens to know exactly where their complaint currently stands.

Possible complaint statuses include:

- Submitted
- Waiting for Review

- Under Review
- Team Assigned
- Work in Progress
- Problem Resolved
- Ticket Closed
- Rejected

Whenever the complaint status changes, the citizen can view the updated status using the Tracking ID.

Complaint Resolution Report

Once the issue has been resolved, the citizen receives a downloadable PDF report containing:

- Complaint Tracking ID
- Complaint category
- Citizen information
- Original complaint description
- Date and time of complaint submission
- Date and time of complaint closure
- Department assigned
- Initial uploaded photograph
- Final photograph after resolution
- Resolution summary
- Total resolution time

If the complaint is found to be invalid, duplicate, or false during verification, the ticket is closed with an appropriate reason recorded in the system.

Administrative Module

The administrative dashboard serves as the central control panel for municipal officials.

Admin Authentication Feature

To secure administrative actions, administrative workflows, and sensitive citizen data, access to the admin dashboard is restricted. Administrators must securely authenticate themselves by logging in with authorized credentials (username/email and password) via a secure authentication system.

Once logged in, administrators can access the active tickets dashboard. Each complaint is assigned a priority using predefined rule-based criteria, considering factors such as:

- Complaint category
- Duration of the issue
- Nearby public facilities (schools, hospitals, markets)
- Accident history
- Public safety impact
- Citizen responses to the questionnaire

Authorized administrators can:

- Review complaint details
- View uploaded images
- Verify complaint authenticity
- Modify complaint priority if necessary
- Assign the complaint to the appropriate department
- Update complaint status
- Add remarks
- Upload photographs after completion of work
- Close the complaint after verification

Throughout the process, every status update is recorded along with timestamps to maintain a complete audit trail.

Public Transparency Portal

To improve transparency and accountability, the system includes a public dashboard that can be accessed by anyone without logging in.

Visitors can view:

1. Active Complaints

The portal displays all currently active complaints, including:

- Tracking ID
- Complaint category
- General location
- Date of submission
- Current status
- Assigned department (if applicable)

This enables citizens to know whether a problem has already been reported before submitting a duplicate complaint.

2. Resolved Complaints

The portal also displays all successfully resolved complaints. Each resolved complaint includes:

- Complaint category
- General location
- Initial complaint image
- Final image after resolution
- Date of submission
- Date of resolution
- Total resolution time
- Resolution summary

This public archive demonstrates the work completed by municipal authorities and promotes accountability.

Overall Workflow

1. A citizen selects a complaint category.
2. The system presents category-specific Yes/No questions.
3. The citizen enters the location of the issue.
4. The citizen uploads photographs.
5. The citizen writes a detailed description.
6. The citizen provides their name and email address.
7. **The system generates and sends a One-Time Password (OTP) to the citizen's registered email address.**
8. **The citizen inputs and verifies the OTP.**
9. **Once verified, the complaint is submitted.**
10. The system generates a unique Tracking ID.
11. A confirmation email with the Tracking ID is sent to the citizen.
12. **An administrator securely logs in using their credentials.**
13. The complaint appears on the administrative dashboard.
14. The system assigns an initial priority using predefined rules.
15. The authenticated administrator reviews the complaint.
16. The administrator approves or rejects the complaint.
17. If approved, the complaint is assigned to the responsible department.
18. Field staff complete the required work.
19. The administrator uploads photographs showing the completed work.
20. The administrator closes the complaint.
21. A detailed PDF resolution report is generated for the citizen.
22. The resolved complaint is published on the public transparency portal.

Key Features

- Secure citizen complaint registration
- **Email-based OTP verification before ticket submission to eliminate spam**
- **Secure admin authentication system for access control**
- Category-specific dynamic complaint forms
- Rule-based priority assignment
- Image upload support (pre and post-resolution)
- Email confirmation with Tracking ID
- Real-time complaint status tracking
- Administrative dashboard for complaint management
- Department assignment and workflow management
- Status history with timestamps
- Automatic PDF resolution report generation
- Public display of active complaints
- Public archive of resolved complaints

- Prevention of duplicate reporting through public visibility
- Complete transparency throughout the complaint lifecycle

Expected Benefits

- Faster reporting of civic issues
- Organized complaint management and accountability
- Reduced paperwork and manual delays
- Improved communication between citizens and authorities
- Secure and controlled system access for administration
- Trustworthy submissions using real-time citizen verification
- Transparent complaint handling process
- Better accountability of municipal departments
- Historical record of resolved civic issues
- Increased public trust through open access to complaint status
- Efficient monitoring of complaint resolution timelines
- Digital documentation of the entire complaint lifecycle

